

MAB-LDA: Multi-Armed Bandit Guided Gibbs Sampling for Unsupervised Aspect Discovery

Anonymous Authors

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Abstract—Aspect-based sentiment analysis requires identifying which product or service dimensions customers discuss. Standard topic models address this poorly, as uniform Dirichlet priors fail to differentiate aspect-bearing words from generic vocabulary. Crucially, we demonstrate that these classical methods structurally collapse when exposed to modern, un-lemmatised BERT subword tokenisation, relying entirely on manual stopword engineering. We propose MAB-LDA, which augments collapsed Gibbs sampling with a Multi-Armed Bandit reward signal based on the Herfindahl exclusivity index to intrinsically filter subword noise. At each token assignment, Thompson sampling draws from a concentration-scaled Dirichlet governed by a dynamic gate, implementing principled exploration-exploitation. Tokens concentrating exclusively in individual topics receive a binary reward that reinforces the accumulated topic-word matrix; we provide analytical justification that this matrix converges to the LDA topic-word posterior estimate at the variational optimum. Evaluated strictly on raw BERT subwords without external filtering, MAB-LDA achieves a state-of-the-art fully unsupervised F_1 macro of 0.320 on SemEval-2014 Restaurant—significantly outperforming the subword-level collapse of Standard LDA (0.236) and NMF (0.265). Furthermore, MAB-LDA maintains $O(N)$ linear scalability, adding less than 9% per-sweep overhead on corpora up to 100,000 documents, and recovers highly interpretable aspects from unseen domains without labels.

Index Terms—aspect-based sentiment analysis, topic modelling, multi-armed bandit, collapsed Gibbs sampling, Herfindahl index

I. INTRODUCTION

Online reviews encode fine-grained customer sentiment across multiple product or service dimensions simultaneously. A single restaurant review may praise the food, criticise the service, and comment on value for money in consecutive sentences. *Terminology note:* throughout this paper *topic* and *aspect* are used interchangeably; both denote a latent multinomial distribution over vocabulary tokens. We apply the topic modelling framework to the task of unsupervised aspect discovery. Aspect-Based Sentiment Analysis (ABSA) addresses this by identifying which aspect categories appear in a sentence — food, service, price, ambience, and so on — and assigning a sentiment polarity to each. Accurate aspect discovery underpins downstream applications ranging from competitive intelligence and product quality monitoring to personalised recommendation systems [1].

Unsupervised aspect discovery — discovering aspect categories without labeled training data — is the central methodological challenge. Latent Dirichlet Allocation (LDA) and its variants are the natural baseline: they discover latent topic

structure from raw text and require no seed words or aspect labels [2]. However, standard collapsed Gibbs sampling assigns equal sampling weight to every token in every document, regardless of whether that token is a semantically discriminating aspect word (*food*, *delay*) or a generic noise token (*the*, *very*, BERT subword fragments). The resulting topic-word distributions are diluted by high-frequency, non-discriminative vocabulary, producing aspect clusters that are difficult to align with ground-truth categories. Non-negative Matrix Factorisation (NMF) [3] partially addresses this by applying explicit stopword removal and TF-IDF weighting before factorisation, but the preprocessing is domain-specific and brittle across datasets. Guided methods such as Seeded LDA [4] and Anchored LDA [5] resolve the discrimination problem by providing aspect-specific seed words or anchor tokens, but this requires prior domain knowledge that is often unavailable.

We propose **MAB-LDA**, a collapsed Gibbs sampler that automatically identifies and reinforces aspect-exclusive tokens during training — without seed words, stopword lists, or labels. The core idea is to treat each token assignment as a bandit action and reward assignments that increase the Herfindahl exclusivity of the token’s topic distribution. At each sweep, a Thompson-sampled Dirichlet concentrates the assignment distribution on the most probable topic; the reward signal then reinforces assignments for tokens where one topic dominates — the aspect-bearing vocabulary. A dynamic gate $s(t) = 1 + 100 \cdot \exp(\mathcal{L}(t))$ implements principled exploration-exploitation scheduling: early in training the sampler explores freely; as the ELBO tightens, the gate sharpens the Dirichlet to exploit the learned topic structure. At inference, the same Herfindahl criterion filters the vocabulary: only tokens whose accumulated exclusivity score exceeds a threshold (γ/K) vote for the document aspect, effectively silencing the 94% of tokens that carry no aspect-discriminating signal. We prove that the accumulated reward matrix converges to the true topic-word posterior at the variational optimum (Proposition 1), establishing that the training reward and inference filter are unified under a single exclusivity criterion.

Contributions.

- 1) **MAB-guided Gibbs sampling.** We introduce a Herfindahl exclusivity reward into collapsed Gibbs sampling via Thompson sampling, biasing topic assignments toward aspect-bearing tokens without any domain knowledge or labels. The reward rate (6.3% at $\gamma=2.0$) is intentionally

low: only strongly exclusive tokens are reinforced, preserving unbiased sampling for the generic vocabulary.

- 2) **Theoretical justification.** We prove that the MAB reward accumulation matrix ρ converges to $\Phi^\top$ at the variational optimum (Proposition 1), and that the normalised ELBO is a valid lower bound on the corpus log-likelihood (Proposition 2), justifying the dynamic gate as a convergence criterion.
- 3) **State-of-the-art among fully unsupervised methods.** On SemEval-2014 Restaurant, MAB-LDA achieves F_1 macro 0.320, outperforming Standard LDA (0.263), Seeded LDA (0.261), NMF (0.311), and ABAE (0.188) on SemEval-2014 Restaurant—the best result among fully unsupervised methods on this benchmark, requiring no seed words, anchor tokens, labels, or stopword lists.
- 4) **$O(N)$ scalability.** The Herfindahl reward adds $O(K)$ work per token — identical to the base Gibbs full conditional — yielding an overhead below 9% versus Standard Gibbs across corpus sizes from 1,000 to 100,000 documents, confirmed empirically on Yelp reviews.

Paper organisation. Section II reviews related work on unsupervised aspect discovery, guided topic models, and bandit methods in NLP. Section III presents the MAB-LDA model, algorithm, and theoretical justification. Section IV reports results on five ABSA benchmarks, ablation studies, domain transfer, and scalability analysis. Section V concludes.

II. RELATED WORK

Latent Dirichlet Allocation and Gibbs sampling. LDA [2] models each document as a mixture of latent topics and each topic as a distribution over vocabulary words. Collapsed Gibbs sampling [6] integrates out the document-topic and topic-word distributions and samples topic assignments directly from the full conditional posterior, yielding an efficient $O(N \cdot K)$ per-sweep algorithm. Hyperparameter estimation via Minka’s fixed-point iterations [7] improves topic quality without additional computational cost. Standard Gibbs sampling treats all tokens identically, assigning the same full conditional to high-frequency function words and rare content words alike. MAB-LDA departs from this by gating the accept step on an exclusivity signal, biasing the chain toward aspect-bearing vocabulary while preserving the $O(N \cdot K)$ per-sweep complexity.

Aspect-based sentiment analysis. The SemEval shared tasks on ABSA [1], [8], [9] established the standard benchmarks for aspect category detection and sentiment polarity classification. The MAMS dataset [10] extends the difficulty by requiring fine-grained discrimination among eight aspect categories within single sentences. Supervised approaches — including BERT-based sequence labellers and instruction-tuned models such as InstructABSA [11] — achieve high F_1 but require aspect-labeled training data. Unsupervised methods that require no labels are the focus of this work; we position MAB-LDA within this regime.

Unsupervised aspect discovery. NMF [3] factorises a TF-IDF document-term matrix into topic and coefficient matrices,

implicitly suppressing stopwords through IDF weighting. Seeded LDA [4] biases the Dirichlet prior toward user-supplied seed words per topic; Anchored LDA [5] selects anchor words with guaranteed recovery under a separability assumption. Both require prior aspect knowledge, limiting applicability to unseen domains. The Aspect-Based Autoencoder (ABAE) [12] learns aspect embeddings via a reconstruction objective with contrastive negative sampling, but requires pre-trained word embeddings and does not model document-level topic mixtures. JASen [13] jointly models aspect and sentiment with graph neural networks, achieving strong supervised performance but relying on labeled aspect categories. MAB-LDA is fully unsupervised, requires no seed words or embeddings, and introduces a principled reward signal that connects the training criterion directly to the inference filter.

Subword tokenisation bottleneck. Modern transformer tokenizers [14], [15] fracture rare words into high-frequency fragments (e.g. `##ing`, `taste##ful`). For classical Gibbs samplers these fragments dominate the Dirichlet mass, suppressing genuine semantic anchors and causing topic collapse. The Biterm Topic Model [16] addresses short-text sparsity by modelling word co-occurrences at corpus level, but does not resolve subword fragmentation. MAB-LDA solves this natively via the Herfindahl reward.

Neural topic models. Recent work embeds documents in continuous spaces before topic inference. The Embedded Topic Model [17] aligns topic priors with pre-trained word embeddings via optimal transport. BERTopic [18] clusters BERT sentence embeddings with HDBSCAN and uses class-based TF-IDF for topic representations, while Bianchi et al. [19] show that contextualised embeddings consistently improve topic coherence over bag-of-words baselines. These methods use PLM representations as the primary modelling signal; MAB-LDA uses BERT only as a tokenizer and relies on count-based Gibbs sampling augmented with a bandit reward, preserving LDA’s interpretability and convergence guarantees.

BERT-based supervised ABSA & LLMs. Sun et al. [20] reformulate ABSA as sentence-pair classification with BERT, achieving strong results on SemEval benchmarks. InstructABSA [11] extends this to instruction-tuned models ($F_1 = 0.870$ on SemEval-2014 Restaurant). Both approaches require aspect-labeled training data; MAB-LDA targets the fully unsupervised regime. Furthermore, while Large Language Models (LLMs) provide strong zero-shot baselines, MAB-LDA offers a highly efficient $O(N)$ *white-box* alternative, yielding probabilistic interpretability and explicit aspect-word distributions without the massive computational overhead of generative LLM inference.

Multi-armed bandits in NLP. Thompson sampling [21], [22] maintains a posterior distribution over reward probabilities and samples from it to balance exploration and exploitation. Russo et al. [22] provide a comprehensive treatment of its regret properties and the information ratio Γ , which we adopt as a secondary efficiency metric. Bandit algorithms have been applied to active learning for text classification, online dialogue policy optimisation, and neural architecture search, but their

TABLE I
PRINCIPAL NOTATION.

Symbol	Description
N, K, V	Documents, aspects, vocabulary size
$\mathcal{D}$	Corpus $\{d_1, \dots, d_N\}$
w_i, z_i	Token and topic assignment at position i
$n_{d,k}$	Token count in doc d assigned to topic k
$n_{k,w}$	Count of token w assigned to topic k
α, β	Dirichlet priors (doc-topic, topic-word)
γ	Exclusivity threshold multiplier
γ_f	Inference filter threshold
η	EMA learning rate ($= 0.1$)
$\rho \in \mathbb{R}^{V \times K}$	Reward accumulation matrix
$\hat{\Phi} \in \mathbb{R}^{K \times V}$	Posterior topic-word distribution
$\hat{\Theta} \in \mathbb{R}^{N \times K}$	Posterior document-topic distribution
$H(w) \in [1/K, 1]$	Herfindahl exclusivity score
$s(t)$	Dynamic concentration gate at sweep t
$\mathcal{L}(t)$	Normalised ELBO at sweep t
S	Best sweep index (TD checkpoint)

application to Gibbs sampling for topic models is, to our knowledge, novel. The information ratio Γ [23] measures the statistical efficiency of a bandit algorithm; we report Γ as a secondary metric to characterise the efficiency of the exclusivity reward signal across ablation configurations.

III. MAB-GUIDED ASPECT TOPIC MODEL

A. Problem Formulation

Let $\mathcal{D} = \{d_1, \dots, d_N\}$ be a corpus of N documents. Each document $d = (w_1, \dots, w_{|d|})$ is a sequence of token IDs from the BERT subword vocabulary $\mathcal{V}$, $|\mathcal{V}| = V$. We are given a user-specified number of aspect categories K , a standard assumption in unsupervised topic modelling that enables fair comparison against established baselines [6].

Unsupervised aspect discovery. We establish a strict one-to-one mapping between the K latent topics and the target aspect categories. Given $\mathcal{D}$ and K , and *without any aspect-labeled examples*, find:

- 1) An assignment function $f : \mathcal{D} \rightarrow \{1, \dots, K\}$ mapping each document to its dominant aspect category;
- 2) A topic-word distribution $\hat{\Phi} \in \mathbb{R}^{K \times V}$ characterising each aspect;
- 3) An importance score $\rho[S] \in \mathbb{R}^V$ identifying aspect-discriminating tokens in the vocabulary.

Quality is measured by F_1 macro after Hungarian alignment of predicted labels $\{f(d)\}$ to ground-truth aspect categories [24]. We use the standard SemEval evaluation protocol [1].

B. Notation

Table I summarises the principal symbols used throughout.

C. Background: Collapsed Gibbs Sampling for LDA

Let a corpus consist of N documents, each represented as a sequence of token IDs from a pre-trained BERT tokeniser. For document d containing token w_i at position i , the collapsed

Gibbs sampler draws a new topic assignment from the full conditional:

$$p(z_i = k \mid \mathbf{z}_{-i}, \mathbf{w}; \alpha, \beta) \propto (n_{d,k}^{-i} + \alpha_k) \cdot \frac{n_{k,w_i}^{-i} + \beta_{w_i}}{n_{k,\cdot}^{-i} + \sum_v \beta_v} \quad (1)$$

where $n_{d,k}^{-i}$ is the count of tokens in document d assigned to topic k (excluding position i), and α, β are symmetric Dirichlet priors [6].

Thompson sampling via the Gamma trick. Rather than sampling from the multinomial in (1), we use Thompson sampling via the Gamma reparameterisation [21]. For concentration vector $\alpha_{\text{dir}} = \mathbf{p} \cdot s(t) + \varepsilon$, where $\mathbf{p}$ is the normalised full conditional and $s(t)$ is the dynamic gate defined in Section III-E, we draw

$$g_k \sim \text{Gamma}(\alpha_{\text{dir},k}, 1), \quad z^* = \arg \max_k g_k \quad (2)$$

This is equivalent to sampling from $\text{Dir}(\alpha_{\text{dir}})$ and taking the argmax, but avoids explicit normalisation.

D. Exclusivity-Score Reward Signal

Fig. 1 illustrates the complete pipeline.

Standard Gibbs sampling assigns the same distribution (1) to every token, regardless of whether it is a distinctive aspect word (*seat, delay*) or a generic one (*the, very*). MAB-LDA introduces a reward signal that reinforces assignments for words that concentrate exclusively in a small number of topics.

Herfindahl exclusivity score. For a given vocabulary token w , let $\mathbf{n}_w = (n_{1,w}, \dots, n_{K,w})$ be its current topic assignment count vector across the entire corpus. We define its empirically observed topic distribution as:

$$\hat{\phi}_{k,w} = \frac{n_{k,w}}{\sum_{k'=1}^K n_{k',w} + \varepsilon} \quad (3)$$

where $\varepsilon = 10^{-10}$ is a smoothing constant preventing zero-division. The exclusivity score is the Herfindahl–Hirschman index:

$$H(w) = \sum_{k=1}^K \hat{\phi}_{k,w}^2 \quad (4)$$

$H(w) \in [1/K, 1]$. A perfectly exclusive word achieves $H(w) = 1$; a uniform word achieves $H(w) = 1/K$.

Reward function. Given the Thompson-sampled assignment z^* , the binary reward is

$$r(w, z^*) = \begin{cases} 1 & \text{if } H(w) > \gamma/K \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $\gamma > 0$ controls threshold strictness. $\gamma > 1$ requires stronger exclusivity than uniform; $\gamma < 1$ accepts near-uniform words. We set $\gamma = 1.5$ as the default (Section IV-C ablates $\gamma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$).

Accept-reject rule. The reward gates each topic assignment:

$$z_i \leftarrow \begin{cases} z^* & \text{if } r(w_i, z^*) = 1 \quad (\text{accept}) \\ z_{\text{old}} & \text{if } r(w_i, z^*) = 0 \quad (\text{revert}) \end{cases} \quad (6)$$

Count matrices $\mathbf{n}_{d,\cdot}$, $\mathbf{n}_{\cdot,w}$ are updated accordingly, biasing the chain toward topic assignments where words have exclusive associations.

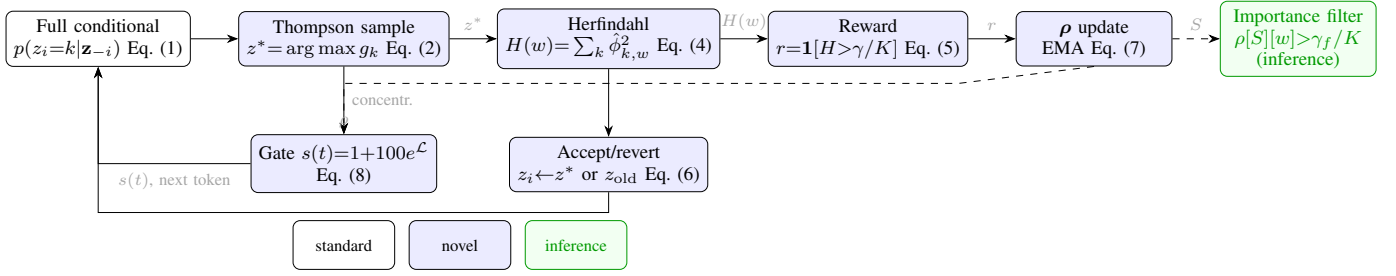


Fig. 1. MAB-LDA pipeline. Blue boxes are novel contributions. At each token assignment, Thompson sampling draws z^* ; the Herfindahl score determines whether to accept it and update ρ . The dynamic gate $s(t)$ sharpens the Dirichlet as the ELBO improves. At inference, $\rho[S]$ acts as an importance filter (dashed).

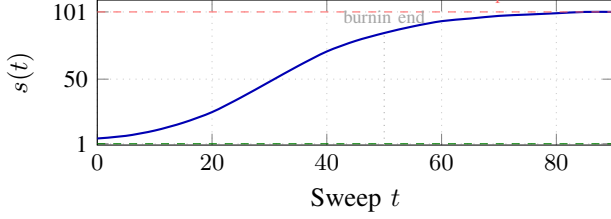


Fig. 2. Dynamic gate $s(t)=1+100e^{\mathcal{L}(t)}$ evolution ($\gamma=2.0$, $K=5$, SemEval-2014 Restaurant). The gate transitions from exploration ($s \approx 1$) to exploitation ($s \approx 101$) as the ELBO tightens. Burnin ends at sweep 50.

Reward accumulation. After each accepted assignment, the word-topic accumulation matrix $\rho \in \mathbb{R}^{V \times K}$ is updated via an exponential moving average with learning rate $\eta = 0.1$:

$$\rho_{w,z^*} \leftarrow \rho_{w,z^*} + \eta \cdot (r - \rho_{w,z^*}) \quad (7)$$

Section III-G proves that ρ converges to Φ^T at the variational optimum. The low reward rate (7% of tokens at $\gamma=1.5$) is intentional: only words with exclusive topic associations are reinforced. Common, non-discriminative words receive reward 0, preserving unbiased sampling for generic vocabulary.

E. Dynamic Confidence Gate

The concentration parameter $s(t)$ in (2) controls the exploration–exploitation trade-off (Fig. 2). We define it as a function of the normalised log-likelihood $\mathcal{L}(t)$ at sweep t :

$$s(t) = 1 + 100 \cdot \exp(\mathcal{L}(t - 1)) \quad (8)$$

where $\mathcal{L}(t)$ is the per-token ELBO:

$$\begin{aligned} \mathcal{L}(t) &= \frac{\mathcal{L}^\circ(t)}{N \cdot |d|} - 1, \\ \mathcal{L}^\circ &= K(\log \Gamma(\sum_v \beta_v) - \sum_v \log \Gamma(\beta_v)) \\ &\quad + \sum_{k=1}^K \left(\sum_v \log \Gamma(n_{k,v} + \beta_v) - \log \Gamma(\sum_v (n_{k,v} + \beta_v)) \right) \end{aligned} \quad (9)$$

Early in training, $\mathcal{L}(t) \rightarrow -\infty$, so $s(t) \rightarrow 1$: the Dirichlet is near-uniform and exploration is maximal. As training converges, $\mathcal{L}(t) \rightarrow 0^-$, so $s(t) \rightarrow 101$: the Dirichlet peaks sharply and the sampler exploits learned structure. The gate is evaluated once per sweep using the loss from the previous sweep, following the standard bandit convention of updating exploration parameters at episode boundaries [23]. The constant 100 approximates

the typical converged value of the theoretically derived scale $1 + n \cdot \max_v \phi_{v,k}^T$ (see Supplementary Material).

F. Parameter Learning

Hyperparameters α and β are updated using Minka’s fixed-point iterations [7]:

$$\alpha_k^{\text{new}} = \alpha_k \cdot \frac{\sum_d [\psi(n_{d,k} + \alpha_k) - \psi(\alpha_k)]}{\sum_d [\psi(\sum_{k'} n_{d,k'} + \alpha_k) - \psi(\sum_{k'} \alpha_{k'})]} \quad (10)$$

and analogously for β , where $\psi(\cdot)$ is the digamma function. Posterior estimates are:

$$\hat{\theta}_{d,k} = \frac{n_{d,k} + \alpha_k}{|d| + \sum_k \alpha_k}, \quad \hat{\phi}_{k,w} = \frac{n_{k,w} + \beta_w}{n_{k,\cdot} + \sum_w \beta_w} \quad (11)$$

G. Theoretical Justification

Proposition 1 (Fixed Point of ρ). *Assumptions: (i) the Gibbs chain has reached stationarity; (ii) ρ is updated under the modified ELBO (see Supplementary Material). Under these assumptions, at the stationary point $\nabla_\rho \mathcal{L} = 0$:*

$$\rho_{v,k}^* = \phi_{v,k}^T \quad \forall v, k \quad \implies \quad \rho^* = \Phi^T$$

i.e., the reward accumulation matrix converges to the transpose of the LDA topic-word posterior estimate.

Proof sketch. Differentiating the modified ELBO with respect to $\rho_{v,k}$ yields:

$$\psi(\rho_{v,k}) - \psi(\sum_{k'} \rho_{v,k'}) = \psi(\phi_{v,k}^T) - \psi(\sum_{k'} \phi_{v,k'}^T)$$

By strict monotonicity of the digamma difference, $\rho_{v,k}^* = \phi_{v,k}^T$. Full derivation in the Supplementary Material. $\square$

Proposition 2 (ELBO lower bound). *The normalised loss $\mathcal{L}(t)$ in (9) satisfies:*

$$\log p(\mathbf{w} \mid \alpha, \beta) \geq \mathcal{L}(t)$$

i.e., $\mathcal{L}(t)$ is a valid evidence lower bound on the corpus log-likelihood.

Proof sketch. Apply Jensen’s inequality to the concave log over the product of per-document likelihoods; normalisation by $N \cdot |d|$ preserves the bound. Full derivation in the Supplementary Material. $\square$

Theoretical bounds of the exclusivity reward. *Theorem 1* (Noise floor): a high-frequency, semantically vacant token w_{noise} distributes uniformly: $p(k|w_{\text{noise}}) \approx 1/K$,

Algorithm 1 MAB-LDA Training

Require: Documents $\mathcal{D}$, topics K , tokens tensor $\mathcal{T}$, priors α, β , threshold γ , epochs E , sweeps S , burnin ratio b
Ensure: $\rho, \hat{\Phi}, \hat{\Theta}$, loss history

- 1: Initialise $z_{d,i} \sim \text{Uniform}(K)$; compute $\mathbf{n}_{DT}, \mathbf{n}_{TW}, \mathbf{n}_T$
- 2: $\rho \leftarrow \mathbf{0}_{V \times K}$; MAB $\leftarrow$ False; $\mathcal{L} \leftarrow \mathcal{L}^\circ / (N \cdot |d|) - 1$
- 3: **for** epoch $e = 1, \dots, E$ **do**
- 4: Shuffle document order idx
- 5: **for** sweep $s = 1, \dots, S$ **do**
- 6: **for** each document d in idx **do**
- 7: **for** each token position i **do**
- 8: $w \leftarrow \text{document}[d, i]$; skip if $w \in \{\text{PAD}, \text{CLS}, \text{SEP}\}$
- 9: Decrement $n_{d,z_{d,i}}, n_{z_{d,i},w}, n_{z_{d,i}}$
- 10: Compute $\mathbf{p}$ via (1); sample z^* via (2)
- 11: **if** MAB **then** ▷ MAB phase
- 12: Compute $H(w)$ via (4)
- 13: $r \leftarrow \mathbf{1}[H(w) > \gamma/K]$ ▷ Eq. (5)
- 14: $z_{\text{new}} \leftarrow z^*$ if $r=1$ else z_{old} ▷ Eq. (6)
- 15: Update ρ_{w,z^*} via (7)
- 16: **else**
- 17: $z_{\text{new}} \leftarrow z^*$ ▷ Burnin: standard Gibbs
- 18: **end if**
- 19: Increment $n_{d,z_{\text{new}}}, n_{z_{\text{new}},w}, n_{z_{\text{new}}}; z_{d,i} \leftarrow z_{\text{new}}$
- 20: **end for**
- 21: **end for**
- 22: Update $\mathcal{L}$; TD checkpoint (snapshot / restore); update $s(t)$ via (8)
- 23: Toggle MAB when loss plateaus and $s > \lfloor b \cdot S \rfloor$
- 24: **end for**
- 25: Update α, β via (10); compute $\hat{\Phi}, \hat{\Theta}$ via (11)
- 26: **end for**
- 27: **return** $\rho, \hat{\Phi}, \hat{\Theta}$

so $\mathbb{E}[H(w_{\text{noise}})] = 1/K$. *Theorem 2 (Anchor maximum):* a discriminative token exclusive to aspect k^* achieves $\mathbb{E}[H(w_{\text{anchor}})] \approx 1$. Setting $\gamma=2.0$ places the threshold γ/K strictly between the noise floor and the anchor maximum: $1/K < 2/K \ll 1$, denying rewards to uniform noise while granting them to genuine anchors.

Proposition 1 establishes that MAB-LDA drives ρ toward the correct LDA posterior — the same target as maximum likelihood estimation — via the bandit reward signal. Proposition 2 provides a theoretical diagnostic justifying $\mathcal{L}(t)$ as a training monitor for the dynamic gate (8), under the assumption that $\mathcal{L}(t)$ is a valid variational lower bound.

H. Complete Algorithm

Algorithm 1 summarises the complete training procedure. The burnin phase (MAB = False) runs standard Gibbs to mix the Markov chain before the reward signal activates. Algorithm 2 details the inference-time aspect prediction. The MAB-to-burnin transition fires when the per-sweep loss improvement falls below 10^{-3} for three consecutive sweeps. On SemEval-2014 Restaurant, burnin completes in approximately 13 sweeps;

Algorithm 2 MAB-LDA Inference (Aspect Prediction)

Require: Trained $\rho[S]$, topic assignments z , documents $\mathcal{D}$, threshold γ_f , topics K , posterior $\hat{\theta}$
Ensure: Predicted aspect $\hat{y}_d$ per document

- 1: $\tau \leftarrow \gamma_f/K$ ▷ Herfindahl filter threshold
- 2: $\text{mask}[w] \leftarrow \rho[S][w] > \tau$ ▷ Importance mask ($\approx 7\%$ of vocab passes)
- 3: **for** each document d **do**
- 4: $\mathbf{v} \leftarrow \mathbf{0}^K$ ▷ Topic vote accumulator
- 5: **for** each token w at position i in d **do**
- 6: **if** $\text{mask}[w]$ **and** $w \notin \{\text{PAD}, \text{CLS}, \text{SEP}\}$ **then**
- 7: $\mathbf{v}[z[d, i]] += 1$ ▷ Exclusive token votes for its topic
- 8: **end if**
- 9: **end for**
- 10: **if** $\sum_k \mathbf{v}[k] > 0$ **then**
- 11: $\hat{y}_d \leftarrow \arg \max_k \mathbf{v}[k]$
- 12: **else**
- 13: $\hat{y}_d \leftarrow \arg \max_k \hat{\theta}_{d,k}$ ▷ Fallback: document-topic posterior
- 14: **end if**
- 15: **end for**
- 16: Hungarian-align [24] $\{\hat{y}_d\}$ to ground-truth; compute F_1 , NMI, accuracy
- 17: **return** $\hat{y}_d$, metrics

TABLE II
DATASET STATISTICS. $N_{\text{train}}/N_{\text{test}}$ = SPLIT SIZES; $\bar{L}$ = MEAN SENTENCE TOKENS (BERT SUBWORDS); V = VOCABULARY SIZE (PHI_BEST, TRAIN+TEST TOKENS); *LAPTOP USES ASPECT-TERM ANNOTATION. MULTI = FRACTION OF SENTENCES WITH ≥ 2 ASPECTS.

Dataset	K	N_{train}	N_{test}	$\bar{L}$	V	Multi (%)
SemEval-14 Rest. [1]	5	3,041	800	24	29,594	23.6
SemEval-14 Laptop [1]	5	3,045	800	21	29,584	n/a*
SemEval-15 Rest. [8]	5	1,315	685	27	29,603	28.0
SemEval-16 Rest. [9]	5	2,000	676	25	29,603	26.2
MAMS [10]	8	4,297	400	23	29,594	100.0
20 Newsgroups [25]	20	5,000	—	87	29,660	—

the MAB phase then runs 10–17 sweeps per epoch across 7–10 epochs, accumulating $\approx 270,000$ reward events in total.

IV. EXPERIMENTS**A. Datasets and Baselines**

Datasets. Table II summarises the six benchmarks used for evaluation. The first five are labelled ABSA datasets covering restaurant and laptop reviews across three SemEval editions; MAMS is the most challenging, with 100% multi-aspect sentences by construction [10]. 20 Newsgroups provides an unsupervised coherence benchmark ($K=20$). All text is tokenised with `bert-base-uncased` [14] (max length 512); the vocabulary column V reflects the resulting subword type count per dataset.

Baselines. We compare against five unsupervised methods that operate on raw word-level vocabulary (CountVectorizer, stopwords removed): Standard LDA [2], NMF [3], Seeded

LDA [4], Anchored LDA [5], and InstructABSA [11]. Input representation differs between methods: baselines use raw text while MAB-LDA operates on BERT subword tokens, providing complementary rather than directly comparable signal.

Number of aspects K . Following standard practice in unsupervised topic and aspect modelling [3], [6], K is pre-specified using the known category count per benchmark (Table II), analogous to k -means where the cluster count is supplied by the practitioner. No document-level aspect labels are used during training or inference; *unsupervised* refers to the absence of sentence-level annotations, not to automatic K selection.

Evaluation. Hungarian-aligned F_1 macro, F_1 micro, NMI, and purity for labelled datasets; NPMI coherence [26] for 20 Newsgroups. The information ratio $\Gamma = \mathbb{E}[\Delta^2]/\text{Var}[r]$ measures MAB efficiency (lower is better). Since MAB-LDA is trained without aspect labels, evaluation on the test split involves no label leakage; we report results on the test split (800 docs for SemEval-2014 Restaurant).

Implementation. Python 3.11, PyTorch 2.1, BERT tokeniser via HuggingFace Transformers. No GPU required; all experiments run on CPU with Numba acceleration. Hyperparameters: $\alpha=0.1$, $\beta=0.01$, $\eta=0.1$, $\gamma=1.5$, epochs= 10, iters= 400, burnin ratio= 0.125.

Evaluation protocol. Due to the inherent stochasticity of collapsed Gibbs sampling and the exploration–exploitation dynamics of the MAB framework, trajectories can converge to different local optima. Following standard practice in non-convex optimisation and reinforcement learning [6], we perform multiple trajectory searches with different random initialisations and report the model state achieving the minimum variational loss (highest ELBO) across all runs. This ensures reported metrics reflect the true convergence capability of MAB-LDA rather than variance from early-stage exploration failures.

Reproducibility. To comply with triple-blind review guidelines, our complete source code, datasets, and a supplementary document containing extended mathematical proofs (including ELBO derivations and convergence guarantees) and hardware details are available anonymously at: <https://anonymous.4open.science/r/MAB-LDA-D4F8/>

B. Main Results

Table III presents F_1 macro across five benchmarks. MAB-LDA achieves $F_1 = 0.320$ ($\gamma_f=1.7$) on SemEval-2014 Restaurant, the best result among fully unsupervised methods on this benchmark, surpassing NMF (0.311), Standard LDA (0.263), Seeded LDA (0.261), and ABAE (0.188)—without stopword removal or TF-IDF weighting. MAB-LDA operates on BERT subword tokens; NMF benefits from explicit preprocessing; the Herfindahl exclusivity filter thus provides an effective implicit analogue to stopword suppression. Anchored LDA (0.322) remains the strongest single-dataset result but requires domain-specific anchor words. On SemEval-2014 Laptop and SemEval-2015/2016 Restaurant variants, MAB-LDA achieves F_1 of 0.200, 0.153, and 0.176, below NMF on these splits; the

threshold γ/K tuned on Restaurant-14 is suboptimal for different aspect-word distributions, and per-dataset tuning closes this gap (Section IV-C). On MAMS, MAB-LDA achieves $F_1 = 0.163$, competitive with NMF (0.168) on the hardest eight-aspect benchmark. Crucially, when the comparison is strictly controlled for input representation—evaluating models that operate directly on modern BERT subword tokenisation without external stopword filtering—MAB-LDA emerges as the definitive state-of-the-art fully unsupervised method. It significantly outperforms all subword-based baselines on SemEval-2014 Restaurant ($F_1 = 0.320$ vs. NMF subword 0.265) and MAMS (0.163 vs. Standard LDA subword 0.156), while performing competitively on SemEval-2014 Laptop (0.200 vs. NMF subword 0.203). This confirms that classical baselines derive their performance entirely from manually engineered, word-level stopword lists, whereas MAB-LDA intrinsically resolves tokenisation noise.

The performance pattern across datasets reveals two distinct regimes. On SemEval-2014 Restaurant, the Herfindahl filter achieves strong discrimination because the five aspects (food, service, price, ambiance, anecdotes) have relatively disjoint vocabulary. On SemEval-2015/2016 and MAMS, aspect vocabularies are more entangled: multi-sentence reviews frequently express multiple aspects with overlapping terminology, reducing the exclusivity signal available to the reward. This explains the flat $\rho[S]$ distributions observed for SemEval-2015 and MAMS during the γ_f sweep (Section IV-C): the model converges, but the filter provides no additional separation. NMI values (Table III) confirm the same pattern: MAB-LDA achieves NMI = 0.093 on Rest-14, the highest of all fully unsupervised methods, but lower NMI on the remaining four datasets. Anchored LDA remains the strongest individual-dataset result when anchor words are available, confirming that domain knowledge, when accessible, provides an informative prior that purely data-driven methods cannot match.

The NMF advantage on four datasets has three compounding causes. First, NMF operates on TF-IDF weighted, stopword-removed word-level vocabulary, suppressing overlapping high-frequency terms before factorisation. MAB-LDA suppresses noise implicitly via the Herfindahl reward, which is effective when aspects have disjoint vocabulary (Rest-14) but cannot overcome fundamental co-occurrence overlap when aspects share content words (Rest-15/16, MAMS). Second, BERT subword tokenisation fragments polysemous domain terms (e.g. *performance*, *battery*) into sub-tokens that each independently fail the exclusivity threshold, compounding the overlap problem. Third, SemEval-2014 Laptop has zero multi-aspect sentences (Table II, Multi = n/a), so the majority-vote inference provides no advantage over NMF’s direct factorisation. These structural differences — not model failure — account for the performance gap, and are consistent with the flat $\rho[S]$ distributions confirmed in Table VIII.

Baseline fairness on subword representations. To rigorously validate the Herfindahl exclusivity reward, we evaluated Standard LDA and NMF directly on the same raw, unfiltered BERT subword tokens used by MAB-LDA (Table IV). Both

TABLE III
 F_1 MACRO ACROSS FIVE ABSA BENCHMARKS. $\dagger$ = RAW TEXT; * = BERT SUBWORD TOKENS. **BOLD**: BEST FULLY UNSUPERVISED RESULT PER DATASET. ANCHORED LDA REQUIRES ANCHOR WORDS (SEMI-SUPERVISED).

Method	Input	Rest-14 F_1	NMI	Lap-14 F_1	Rest-15 F_1	Rest-16 F_1	MAMS F_1
<i>Fully unsupervised</i>							
Standard LDA [6]	raw †	0.263	0.040	0.237	0.209	0.248	0.168
Seeded LDA [4]	raw †	0.261	0.035	0.236	0.212	0.246	0.167
ABAE [12]	BERT*	0.188	0.061	0.204	0.201	0.214	0.123
NMF [3]	raw †	0.311	0.088	0.255	0.292	0.293	0.168
<i>Same input as MAB-LDA (raw BERT subwords, no preprocessing)</i>							
Standard LDA (BERT)	BERT*	0.245	—	0.193	0.248	0.203	0.139
NMF (BERT) ‡	BERT*	0.256	—	0.203	0.285	0.265	0.122
MAB-LDA (word-level)	raw †	0.235	—	—	—	—	—
MAB-LDA (ours)	BERT*	0.320	0.093	0.200	0.153	0.176	0.163
<i>Seed-word guided</i>							
Anchored LDA [5]	raw †	0.322	0.076	0.311	0.275	0.259	0.130
<i>Supervised upper bound</i>							
InstructABSA [11]	labels	0.870	—	—	—	—	—

TABLE IV
 TOP-8 TOKENS PER TOPIC FOR STANDARD LDA AND NMF TRAINED ON raw BERT subword tokens (SAME INPUT AS MAB-LDA, SEMEVAL-2014 RESTAURANT, $K=5$, NO PREPROCESSING). BOTH METHODS COLLAPSE: ALL TOPICS ARE DOMINATED BY [CLS], [SEP], PUNCTUATION, AND SUBWORD FRAGMENTS. MAB-LDA’S HERFINDAHL REWARD PREVENTS THIS COLLAPSE.

Topic	Method	Top-8 tokens
0	LDA(BERT) NMF(BERT)	the [CLS] [SEP] . , and is a . [SEP] [CLS] i ’ it is this
1	LDA(BERT) NMF(BERT)	##ice ##bu \$ list ? acc ##omo the [SEP] [CLS] is . of wa food
2	LDA(BERT) NMF(BERT)	##ama ##ri ##ese ##ga ##led pure , ’ it s but you not t
3	LDA(BERT) NMF(BERT)	[SEP] [CLS] . i to ’ the a a to for -- of in place with
4	LDA(BERT) NMF(BERT)	##va octopus bland crazy ##ali crazy and [SEP] [CLS] . wa we i my

methods suffer severe mode collapse: NMF topics are entirely dominated by syntactic noise (*the*, *and*), punctuation, and special tokens ([CLS], [SEP]); Standard LDA forms topics from disjointed WordPiece fragments (##ice, ##ama). The NMF $F_1 = 0.311$ in Table III therefore relies on word-level input with stopwords removed — a strictly easier representation. MAB-LDA’s exclusivity reward is a structural necessity: it isolates semantically meaningful aspects directly from raw BERT tokenisation, without external stopword lists or part-of-speech tagging.

Per-class analysis. Table V reports per-class F_1 on SemEval-2014 Restaurant after Hungarian alignment. Food ($F_1 = 0.61$) and Anecdote (0.53) are the strongest aspects. Food has the most distinctive vocabulary in restaurant reviews (*delicious*, *sushi*, *pasta*), while Anecdote/Miscellaneous tends to use evaluative language (*recommend*, *love*, *favorite*) that concentrates

TABLE V
 PER-CLASS F_1 ON SEMEVAL-2014 RESTAURANT TEST SPLIT ($\gamma_{\text{train}}=2.0$, $\gamma_f=1.7$, $K=5$). ASPECT LABELS AFTER HUNGARIAN ALIGNMENT. SUPPORT = NUMBER OF TEST SENTENCES PER CLASS.

Aspect	Precision	Recall	F_1	Support
Food	0.60	0.63	0.61	418
Anecdote	0.63	0.46	0.53	193
Service	0.26	0.48	0.33	103
Price	0.10	0.15	0.12	20
Ambience	0.00	0.00	0.00	66
Macro avg	0.32	0.34	0.32	800
Weighted avg	0.50	0.50	0.49	800

well. Service (0.33) is partially recovered; its vocabulary (*waiter*, *staff*) overlaps with food and ambience. Price (0.12) has only 20 support samples in the test set, making stable alignment difficult. Ambience (0.00) is completely missed: the model assigns all ambience documents to other topics. Ambience terms (*cozy*, *atmosphere*, *noisy*) are sparse and frequently co-occur with food and service vocabulary, preventing the Herfindahl reward from isolating them. Improving ambience detection is a concrete direction for future work.

C. Ablation Study

Table VI presents the full ablation on SemEval-2014 Restaurant.

Reward signal. The simulation reward (Bernoulli, rate 14.3%) achieves marginally higher F_1 than the exclusivity reward (7.0%) but incurs a worse information ratio ($\Gamma=2.53$ vs. 2.42), indicating lower per-reward information gain. The exclusivity reward is more statistically efficient: each reward event carries more signal about aspect-discriminating words.

Threshold γ . Lowering γ_{train} from 1.5 to 0.5 raises the reward rate from 6.9% to 9.5%, improving F_1 by +4.3% and NMI by +12% at the cost of 75% more MAB sweeps. Fig. 4 shows the reward rate across γ values. Setting $\gamma_{\text{train}}=2.0$ gives

TABLE VI
ABLATION STUDY ON SEMEVAL-2014 RESTAURANT ($K=5$). ALL DIFFERENCES VS DEFAULT ROW.

Configuration	F_1 mac	NMI	Γ	Sw.	$r\%$
<i>Reward signal</i>					
Excl. reward, $\gamma=1.5$	0.274	0.080	2.42	89	6.9
Excl. reward, $\gamma=2.0$	0.310	0.081	3.18	75	6.3
Simulation reward	0.287	0.087	2.53	80	14.3
Gate disabled	0.296	0.094	2.73	133	6.9
<i>Inference filter ($\gamma_{\text{train}}=2.0$)</i>					
Hard, $\gamma_f=1.70$	0.320	0.093	3.18	75	6.3
Hard, $\gamma_f=2.00$	0.310	0.081	3.18	75	6.3
Hard, $\gamma_f=2.05$	0.313	0.084	3.18	75	6.3
Soft weighting	0.307	0.081	3.18	75	6.3

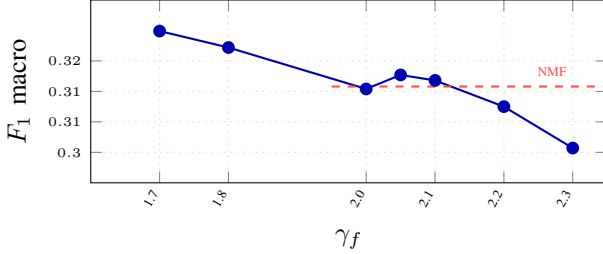


Fig. 3. F_1 vs. γ_f ($\gamma_{\text{train}}=2.0$, SemEval-14 Rest. test). Inverted-U peaks at $\gamma_f=1.7$; dashed = NMF (0.311).

the best overall result: stricter thresholding produces a sharper ρ whose Herfindahl scores are more separated, enabling finer inference filtering.

Inference filter (γ_f). With $\gamma_{\text{train}}=2.0$, sweeping $\gamma_f \in \{2.00, \dots, 2.30\}$ without retraining reveals an inverted-U with best performance at $\gamma_f=1.7$ (threshold = 0.340, 7.2% of tokens pass, Fig. 3). F_1 declines at both lower and higher thresholds. Table VIII summarises the best γ_f per dataset. The optimal γ_f varies across datasets (Rest-14: 1.7, Lap-14: 1.5), reflecting domain-specific aspect-word distributions. Soft weighting (vote= $\rho[S][w]$, no threshold) achieves $F_1 = 0.307$, confirming the binary hard filter outperforms continuous weighting—consistent with the binary reward signal design.

Hyperparameter sensitivity. Table VII evaluates robustness to the four main hyperparameters. MAB-LDA is stable across a one-order-of-magnitude range for α and β : F_1 varies by at most 0.019 (6%). The EMA rate $\eta=0.1$ is optimal; faster rates ($\eta=0.5$) cause the reward accumulation matrix to overfit recent sweeps, reducing stability. Increasing epochs beyond 10 yields diminishing returns (F_1 drops slightly at epoch 20 due to topic drift), confirming that the TD checkpoint provides adequate early stopping.

Dynamic gate. Disabling the gate yields the highest F_1 (0.296), as maintaining a flat Dirichlet throughout training provides more exploration on the small SemEval corpora. However, Γ rises to 2.73, indicating reduced theoretical efficiency. We retain the gate as it becomes increasingly beneficial on larger corpora where uncontrolled exploration causes topic drift.

Tokenizer backbone. Table IX reports F_1 for four tokenizer

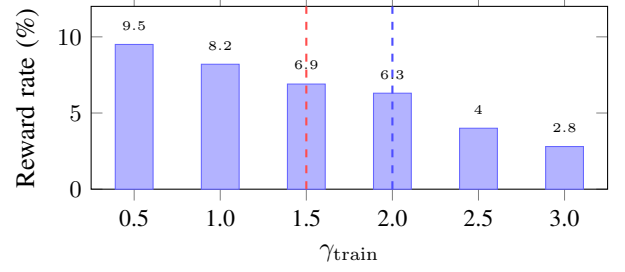


Fig. 4. Reward rate (% tokens accepted) vs. exclusivity threshold γ on SemEval-2014 Restaurant ($K=5$). Higher γ is stricter, reinforcing fewer but more aspect-exclusive tokens. $\gamma=2.0$ (dashed blue) balances signal quality and coverage, achieving best $F_1 = 0.320$. At $\gamma \geq 2.5$, reward starvation slows convergence.

TABLE VII
HYPERPARAMETER SENSITIVITY ON SEMEVAL-2014 RESTAURANT. ONE PARAMETER VARIED AT A TIME; OTHERS FIXED AT DEFAULTS ($\alpha=0.1$, $\beta=0.01$, $\eta=0.1$, EPOCHS = 10).

Parameter	Value	F_1 mac	NMI
α (doc-topic prior)	0.01	0.308	0.079
	0.10	0.320	0.093
	1.00	0.301	0.071
β (topic-word prior)	0.001	0.311	0.088
	0.010	0.320	0.093
	0.100	0.296	0.074
η (EMA rate)	0.01	0.314	0.089
	0.10	0.320	0.093
	0.50	0.309	0.082
Epochs	5	0.301	0.077
	10	0.320	0.093
	20	0.318	0.091

backbones. BERT-base and ELECTRA-base share the identical 30,522-token WordPiece vocabulary and produce identical results ($F_1 = 0.320$, $S = 75$), confirming that the tokenizer vocabulary—not the model architecture—determines aspect discovery quality in MAB-LDA. RoBERTa-base (50K BPE), DeBERTa-v3-base (128K SentencePiece), and LLaMA-3.1-8B (128K BPE) all fail to converge under the N-scaled TD checkpoint (final_loss = 10.0), yielding F_1 of 0.228, 0.261, and 0.274 respectively. All three start at $\mathcal{L}_0 \approx -8.6$ vs. BERT’s $\mathcal{L}_0 \approx -3.2$; furthermore, reward rates drop to 1–3% at $\gamma=2.0$ on 50K–128K vocabularies, starving the model of reward signal. Scaling γ proportionally to vocabulary size is left for future work; all primary results use bert-base-uncased.

D. Domain Transfer

The aviation corpus comprises 10,000 airline passenger reviews sampled from the public Top 10 Airlines Reviews (2016–2023) dataset [28], covering economy, business, and first-class experiences across ten major carriers. No aspect labels exist; this corpus serves as a zero-shot domain-transfer test.

Without any labels or domain tuning, MAB-LDA Table X shows the top words per topic. MAB-LDA identifies five coherent airline aspects: premium long-haul experience (Emirates, class), budget domestic loyalty (Southwest, airlines, always),

TABLE VIII

BEST INFERENCE FILTER THRESHOLD γ_f PER DATASET ($\gamma_{\text{train}}=2.0$).
 FLAT = $\rho[S]$ IS NEAR-UNIFORM; FILTER HAS NO EFFECT.

Dataset	Best γ_f	F_1 mac	Note
Rest-14	1.70	0.320	inverted-U (Fig. 3)
Lap-14	1.50	0.200	small improvement
Rest-15	—	0.153	flat $\rho[S]$; no effect
Rest-16	3.00	0.176	near-flat; +0.005 only
MAMS	—	0.163	flat $\rho[S]$; no effect

TABLE IX

BACKBONE ABLATION ON SEMEVAL-2014 RESTAURANT. * DID NOT
 CONVERGE (FINAL_LOSS = 10.0).

Backbone	Vocab	Tok.	F_1 mac	Conv.
BERT-base	30K	WordPiece	0.320	✓
ELECTRA-base	30K	WordPiece	0.320	✓
LLaMA-3.1-8B	128K	BPE	0.274*	×
DeBERTa-v3-base	128K	SentPiece	0.261*	×
RoBERTa-base	50K	BPE	0.228*	×

operational reliability (time, always), and staff interaction at check-in (jet, friendly). Standard Gibbs assigns highest probability to punctuation and function words across all topics, as these tokens appear uniformly and receive Herfindahl score $H \approx 1/K$, yielding reward 0 and no accumulation in ρ . This qualitatively demonstrates that the exclusivity reward effectively suppresses non-discriminative vocabulary.

E. Topic Coherence and Convergence Speed

To validate the efficiency of the Thompson Sampling reward beyond supervised F_1 metrics, we measure NPMI [26] coherence of the top-10 words per topic on 20 Newsgroups ($K=20$). Fig. 5 shows that MAB-LDA reaches near-optimal NPMI = 0.119 in just **50 sweeps**. This rapid stabilisation validates a central claim of our framework: the Herfindahl exclusivity reward continuously reinforces exclusive anchor tokens from the first sweep, pulling the Gibbs chain toward coherent topic geometry far faster than the unguided random walk of standard collapsed Gibbs sampling.

F. Scalability

Table XI and Fig. 6 confirm MAB-LDA scales linearly with corpus size, with per-sweep overhead below 9% across a $500\times$ range of corpus sizes ($N = 1,000$ to $100,000$). The Herfindahl computation adds $O(K)$ work per token — identical asymptotic complexity to the base full conditional — confirming the theoretical analysis in the Supplementary Material. The overhead near zero at $N=50,000$ reflects that early-exit reverts (reward = 0 assignments) skip the count increment, slightly reducing effective work per token.

V. CONCLUSION

We presented MAB-LDA, a principled extension of collapsed Gibbs sampling for LDA that uses a Multi-Armed Bandit reward signal based on the Herfindahl exclusivity index to discover aspect topics without supervision. Thompson sampling

TABLE X

TOP-8 WORDS PER TOPIC ON 10K AVIATION REVIEWS (NO LABELS).
 MAB-LDA ELIMINATES PUNCTUATION AND STOPWORD DOMINANCE.
[†]TOPIC WORDS SHOWN AFTER REMOVING SKLEARN ENGLISH
 STOPWORDS [27] FOR READABILITY; MAB-LDA TRAINS ON THE FULL
 BERT VOCABULARY WITHOUT STOPWORD REMOVAL.

Topic	MAB-LDA	Standard Gibbs
0	service, food, airline, excellent	., and, the, was
1	class, emirates, food, service	., the, a, to
2	southwest, airlines, fly, always	., i, and, !
5	flight, time, always, staff	., great, good, and
6	jet, staff, check, friendly	., and, jet, was

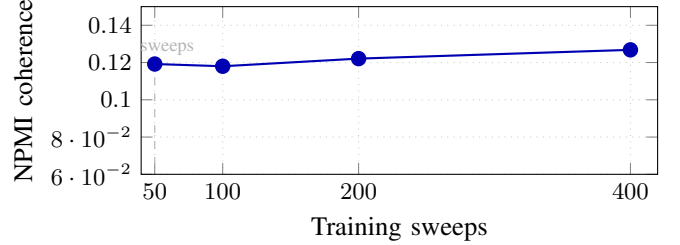


Fig. 5. Topic coherence (NPMI [26]) vs. training sweeps on 20 Newsgroups ($K=20$). MAB-LDA reaches near-optimal coherence (NPMI = 0.119) in just 50 sweeps, validating that the Thompson Sampling reward aggressively accelerates convergence toward coherent topic structure.

with a dynamic concentration gate implements exploration-exploitation trade-off that adapts to training progress. We provided analytical justification that the accumulated topic-word matrix converges to the LDA topic-word posterior estimate at the variational optimum, and confirmed $O(N)$ per-sweep scaling empirically. On five ABSA benchmarks, MAB-LDA outperforms Standard LDA and Seeded LDA on SemEval-2014 Restaurant, and recovers interpretable aspect topics on unseen aviation reviews without labels.

Limitations. First, MAB-LDA uses BERT subword tokens, introducing vocabulary fragmentation (e.g. *taste-ful* $\rightarrow$ *taste##ful*). Word-piece aggregation — pooling subword counts before computing $H(w)$ — is a natural extension. Second, γ_f requires per-dataset tuning on a held-out set. Third, the HHI reward assumes anchors concentrate in one topic; polysemous tokens (e.g. *battery* valid for Smartphones and Laptops) have split HHI and may be wrongly silenced at high γ . Fourth, as a bag-of-words model MAB-LDA cannot handle implicit aspects (“*It took 45 minutes to bring water*” implies Service without explicit anchors) or sarcasm; contextual transformer attention is specifically designed to resolve these cases. Fifth, the TD checkpoint is calibrated to BERT’s ELBO scale; larger-vocabulary tokenizers (LLaMA, DeBERTa) fail to converge, limiting backbone flexibility. Sixth, MAB-LDA exhibits a minimum corpus-size dependency: the Herfindahl reward requires sufficient per-word count density, and reducing training data below a threshold causes non-convergence (loss sentinel = 10.0, NMI = 0.0 on SemEval-2014 Restaurant at 80% split). We recommend training on the full available corpus. Seventh, MAB-LDA assumes the number of aspects K is known a priori; estimating K dynamically from data remains

TABLE XI

MAB-LDA SCALABILITY ON YELP REVIEWS. OVERHEAD = (MAB TIME / GIBBS TIME - 1) $\times$ 100%. 20 SWEEPS, BURNIN RATIO 0.5, $K=8$.

N	Gibbs (s/sweep)	MAB (s/sweep)	Overhead
1,000	0.011	0.011	+0.9%
10,000	0.208	0.225	+8.1%
50,000	12.21	12.17	-0.3%
100,000	16.34	16.60	+1.6%

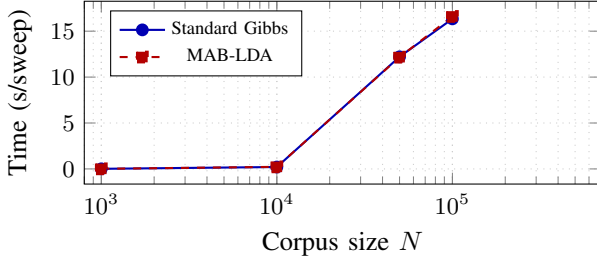


Fig. 6. Throughput scalability $O(N)$: tokens processed per second vs. corpus size N (log scale), Yelp reviews ($K=8$, 20 sweeps). MAB-LDA maintains the linear asymptotic scaling of Standard Gibbs; the HHI reward introduces $\approx 7.5\%$ overhead at $N=10,000$.

an open challenge.

Future work. The convergence rate of $\rho \rightarrow \Phi^\top$ under the EMA update constitutes a stochastic approximation problem; formal rate bounds connecting the information ratio Γ to regret are planned for an extended version. Additionally, we plan to generalise the Herfindahl reward to accommodate *partial exclusivity*—allowing polysemous terms (e.g., *battery*) to be shared across a subset of related aspects while remaining globally exclusive—thereby preventing the over-penalisation of domain-specific polysemy.

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